

Heart Disease Prediction

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Introduction

This report presents the analysis and modeling pipeline applied to a heart disease dataset containing 224 training samples and 16 features, including clinical measurements, demographic information, and lifestyle factors. The target variable, Heart Disease (Yes/No), was balanced across both classes, making standard classification metrics directly applicable.

The work was structured into four main stages: exploratory data analysis and visualization, data preprocessing, and model training and evaluation. The goal was to identify the features most strongly associated with heart disease and build a reliable classifier capable of predicting its presence. Lastly, a simple web interface was also developed using NiceGUI to allow interactive prediction using the trained model.

A. Analysis and Visualizations

1. Target Variable Distribution: Showed that there is no class imbalance, which means that model accuracy then is a reliable metric and we will not use special techniques like SMOTE.

2. Categorical Features vs. Heart Disease

- **Gender:** The countplot showed that most females in the dataset have heart disease, while most males do not, so gender can be a strong predictor
- **Work Type:** People working in the private sector had the highest count of heart disease cases. Children had the lowest. However, work type showed the weakest association with the target in the Cramér's V analysis, so it was retained only because of the small dataset size.
- **Smoking Status:** Former smokers had the highest percentage of not having heart disease, while current smokers had the highest percentage with heart disease. This confirms smoking as a behavioral risk factor.
- **Chest Pain Type:** Chest pain type 4 was strongly associated with heart disease, while type 2 was more common among those without it. This aligns clinically that type 4 (asymptomatic) is the most dangerous.

3. Numerical Features

- **Age Distribution:** The histogram showed Age is near-normally distributed, concentrated between 40–65 years.
- **Age vs. Max HR (scatter plot):** People aged 55–70 were more likely to have heart disease; those aged 30–55 generally don't have disease. This shows that both age and heart rate together are informative.
- **Boxplot (numerical features):** Used to visually check for outliers across all numerical variables.

4. Correlation Analysis

- **Categorical Correlation via Cramér's V:** Used Cramér's V to quantify association between categorical features and the target. **Thallium** had the strongest association, while **work_type** had the weakest.
- **Pearson Correlation with Target:** Plotted each numerical feature's correlation with Heart Disease. The key findings: **Number of vessels fluro** and **Max HR** were most correlated with the target; **Cholesterol** was least correlated.
- **Correlation Heatmap (numerical vs numerical):** Showed pairwise Pearson correlations among numerical features. No severe multicollinearity was detected.

5. Final Insight

The EDA revealed that Thallium scan results, chest pain type, number of fluoroscopy vessels, and maximum heart rate are the strongest indicators of heart disease in this dataset, consistent with established clinical knowledge. Demographic factors like work type and smoking status showed weaker but still meaningful associations. These findings guided both feature encoding decisions and model interpretation."

B. Data Preprocessing

1. Dropping ID column

2. Dealing with Nulls

First, I needed to check which features contain null values.

"Age" had 4 in train data & 2 in test data.

“Gender” had 6 in train data & 0 in test data.

“work_type” had 7 in train data & 2 in test data.

“smoking_status” had 5 in train data & 1 in test data.

- **Dealing with Nulls for Numerical Features:** Since “Age” is a numerical column, the best or the most common case is to fill the nulls with (the most common age in the train data set) in both train data and test data.

Found the median of age in the train dataset and saved it into a variable, used that variable to fill nulls in the “Age” column in the train dataset & test dataset.

- **Dealing with Nulls for Categorical Features:** Since “Gender”, “work_type” and “smoking_status” are categorical columns, the best or the most common case is to fill their nulls with (the most common value in the train data set) in both train data and test data.

Found the mode for each of these features in the train dataset and saved them in separate variables, used each variable to their corresponding feature to fill their nulls in the train dataset & test dataset.

3.Dealing with Duplicates

The train dataset had 9 duplicates and test data had 1.

Dropped all duplicates, to avoid overfitting and inaccurate training.

4.Dealing with Infrequent Categories

Looked into the unique categories of categorical columns and found “Never_worked” and “children” in the “work_type” column.

Dropped the rows associated with these two rare values.

5.Feature Encoding

The categorical features are “Gender”, “Heart Disease”, “Thallium”, “work_type” and “smoking_status”.

Note: “Thallium” column is already encoded with unique values [‘3’, ‘6’, ‘7’].

But the difference here in numbers doesn't mean anything. The gap between 3 and 6 and 3 and 7 or even 6 and 7 means nothing, they are just categories. So, for better accuracy and performance of the model, they should be encoded.

- **One-Hot Encoding:** Encoded "work_type", "smoking_status" and "Thallium" with one-hot encoding.
- **Label Encoding:** Encoded "Gender", "Heart Disease". As they have only two labels.

6.Handeling Outliers

After the analysis and the box-plot presented above, the features that have extreme values seen in the figure are "Cholesterol" and "BP".

Calculated the quartiles (Q1,Q2,Q3) and the IQR to find lower bounds & upper bounds for each of "Cholesterol" and "BP".

Then clipped the both of them with their lower and upper bounds.

7.Feature Scaling

Used MinMax Scaler to keep the values of numerical features between 0-1. For it to be suitable for the Logistic Regression Model.

Note: Any change happened on the train dataset was done on the test dataset using the fit on train dataset to prevent data leakage.

C. Model Selection and Hyperparameters Tuning

Model	Best Hyperparameters	Train Acc.	Test Acc.	Gap	Note
Decision Tree	criterion=entropy, max_depth=4, min_samples_leaf=8, min_samples_split=20	86.10%	84.91%	1.19%	Baseline had a 20.75% gap — reduced by 19.56% while improving test acc. by 5.66%.
Random Forest	criterion=gini, max_depth=None, min_samples_leaf=6, min_samples_split=5, n_estimators=75	90.00%	86.00%	4%	Mild overfitting expected on small datasets. Generalization remains strong.
SVM	C=1, kernel=linear	85.00%	84.00%	1%	Most balanced result. C=100 linear overfits; RBF at C=100 reaches 99% train / 75% test (24% gap).
KNN	K=13, distance=Manhattan	85.00%	83.00%	2%	weights='distance' caused 100% train acc. (severe overfitting). K=13 stabilized the model.
Logistic Regression	C=10, solver=liblinear	85.20 %	84.90%	0.3%	— best generalization of all models. C=0.01 underfits; lbfgs gives near-identical results.
XGBoost	n_estimators=200, max_depth=3, learning_rate=0.01; threshold lowered to 0.40	92.38%	88.68%	3.7%	Highest overall accuracy. Threshold 0.50→0.40 improved recall from 0.77→0.82 with minor precision drop (0.94→0.90).

D. Final Model

After evaluating all models, XGBoost with GridSearchCV achieved the highest test accuracy of 88.7% with a macro F1-score of 0.88, outperforming

Logistic Regression (84.9%), SVM (84.9%), KNN (83%), Decision Tree (84.9%), and Random Forest (86.8%). The best hyperparameters found were: {'colsample_bytree': 1.0, 'learning_rate': 0.01, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.8} The model showed strong precision of 0.94 for detecting heart disease cases and a recall of 0.97 for healthy cases, making it reliable in both directions.

E. GUI and Model Integration

A web-based interface was developed using NiceGUI to allow interactive heart disease prediction without requiring any coding knowledge. The interface collects all 15 patient features through labeled input fields, organized into four sections: Basic Information, Heart Symptoms, Advanced Tests, and Additional Information.

All encoders and the trained model were serialized using joblib and loaded at application startup, ensuring the inference pipeline is identical to the training pipeline. The preprocessed input is then passed to the model, and the prediction result — either "Heart Disease Detected" or "No Heart Disease" — is displayed directly on the screen. The form can be reset to accept a new patient entry without restarting the application.